Mathematical Thinking

Module 1: Introduction to Mathematical Thinking

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目录

1 单元概览

本单元是课程的入门部分，旨在帮助学习者建立正确的数学思维模式。与传统的”学习公式 → 套用解题”不同，本课程强调**理解数学的本质**——数学不仅仅是计算工具，更是一种**精确、严谨、抽象**的思维方式。

核心目标

- 从”学校数学”过渡到”大学数学”的思维模式
- 理解数学是研究**模式**（patterns）的学科
- 掌握精确语言（precise language）在数学中的重要性
- 学会使用逻辑连接词和量词进行严格推理

2 课程讲义

2.1 1. 第-0-讲-欢迎辞

观看视频：第-0-讲-欢迎辞

内容摘要: 1 Hello, I’m Keith Devlin. 2 Welcome to this online course on mathematical thinking. 3 The goal of the course is to help you develop a valuable mental ability, 4 a powerful way of thinking that people have developed over 3,000 years. 5 What I want to do today is get you ready for the course and 6 tell you a little bit about the way the course will work. 7 I’m doing this because for 8 most of you, this will be a very different perspective on what mathematics is. 9 Apart from the final two lecture...

关键帧截图:

时间戳 29s —concept$_{s}hift$

时间戳 44s —concept$_{s}hift$

补充材料 **01__ 第-0-讲-欢迎辞 __Supplement___Set__Theory__.pdf**
KEITH DEVLIN: Introduction to Mathematical Thinking SUPPLEMENT 1 Elements of Set Theory NOT REQUIRED UNTIL LATER IN THE COURSE The later parts of this course assume you (the student) are familiar with basic set theory. This brief document summarizes what is required. Although you won’t need this mat...

01__ 第-0-讲-欢迎辞 __Background__Reading.pdf
KEITH DEVLIN: Introduction to Mathematical Thinking BACKGROUND READING 1 What is mathematics? For all the time schools devote to the teaching of mathematics, very little (if any) is spent trying to convey just what the subject is about. Instead, the focus is on learning and applying various procedur...

2.2 2. 第-1-讲-介绍材料

观看视频：第-1-讲-介绍材料

内容摘要：1 [BLANK AUDIO] I'm going to start off with a question. 2 What is mathematics? 3 That might seem strange given you've probably spent several years being 4 taught math. 5 But for all the times schools devote to the teaching of mathematics, very little, 6 if any is spent trying to convey just what the subject is about. 7 Instead, the focus is on learning and 8 applying various procedures to solve math problems. 9 Well that's a bit like explaining soccer by saying it's a series of maneuvers you 10 ...

关键帧截图：

2.3 3. 第-2-讲-逻辑组合器

观看视频：第-2-讲-逻辑组合器

内容摘要：1 Welcome to the second lecture. 2 I hope you're making progress with the first assignment. 3 You should not expect to solve all the problems in an assignment in 4 a single session. 5 Or even before the next lecture. 6 What you should do before the next lecture is attempt each question. 7 That's what I mean when I say, complete the assignments. 8 Remember, the goal of this course is to acquire a certain way of thinking, 9 not to solve problems by a given deadline. 10 The only way to develop a ne...

关键帧截图：

2.4 4. 作业-1-的教程

观看视频：作业-1-的教程

内容摘要：1 We're in assignment one, most of the questions are just there for 2 you to think about and to discuss with your fellow students. 3 And I'm going to leave you to do that and sort them out on your own, 4 discuss them on the forums or with whatever group you've put together. 5 And please do put together a group. 6 What I will do is say a little bit about question 8. 7 Well, question 8 is related to Euclid's proof that there are infinitely many 8 primes. 9 Another crucial point in that proof, 10 w...

关键帧截图：

时间戳 31s —concept$_shift$

2.5 5. 作业-2-的教程

观看视频：作业-2-的教程

内容摘要：1 Okay, let's see what we've got. 2 Well number one has been done for you. 3 Number two, the simple way to write that is to 4 say 7 less than or equal to p less than 12. 5 The next one we would write as 5 6 less than x less than seven. 7 The next one, well if x, let me see, 8 if x is less than 4 then it's automatically less than 6. 9 So the second conjunct here is superfluous, 10 we could just write that as x less than 4. 11 What about the next one? 12 Well y squared less than 9 means, let me se...

关键帧截图：

时间戳 18s —concept$_shift$

时间戳 27s —concept$_shift$

3 核心概念详解

3.1 逻辑连接词（Logical Connectives）

逻辑连接词是组合命题的基本运算，是数学推理的基石：

五种基本逻辑连接词

1. **合取（AND）**：$P \wedge Q$ —当且仅当 P 和 Q 都为真时为真
2. **析取（OR）**：$P \vee Q$ —当 P 或 Q 至少一个为真时为真
3. **否定（NOT）**：$\neg P$ —将 P 的真值反转
4. **条件（IF...THEN）**：$P \Rightarrow Q$ —仅当 P 真且 Q 假时为假
5. **双条件（IFF）**：$P \Leftrightarrow Q$ —P 和 Q 同真或同假时为真

3.2 量词（Quantifiers）

量词用于表达" 对所有" 或" 存在某个" 的数学陈述：

两个核心量词

- **全称量词**: $\forall x\, P(x)$ —" 对所有 x，P(x) 成立"
- **存在量词**: $\exists x\, P(x)$ —" 存在某个 x 使得 P(x) 成立"

量词否定规则（德摩根律）:

$$\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$$

$$\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$$

3.3 集合论基础（Set Theory Basics）

根据补充材料，集合论是后续课程的数学语言基础：

- **集合表示**: $\{1, 2, 3\}$ 或 $\{x \in N \mid x < 4\}$
- **空集**: $\emptyset$ —不含任何元素的集合
- **属于关系**: $x \in A$ 表示 x 是集合 A 的元素
- **子集**: $A \subseteq B$ 表示 A 的所有元素都在 B 中

4 习题讲解

练习题

1. **量词否定**：写出 $\forall x\,(x>0 \Rightarrow x^2>0)$ 的否定形式。【来源：第二讲】
2. **德摩根律**：用真值表证明 $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$。【综合：第一讲-第二讲】
3. **形式化翻译**：将”每个正数都有平方根”翻译为形式逻辑表达式。【来源：第二讲】
4. **素数无穷**：解释欧几里得证明素数无穷的核心思路。【来源：作业 1 教程】

参考答案

1. 否定形式：$\exists x\,(x>0 \wedge x^2 \leq 0)$

   解析：全称量词变存在量词，条件句 $P \Rightarrow Q$ 的否定是 $P \wedge \neg Q$。

2. 构造 4 行真值表：

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P \vee \neg Q$
T	T	T	F	F
T	F	F	T	T
F	T	F	T	T
F	F	F	T	T

   两列结果完全一致，故等价。

3. $\forall x\,(x>0 \Rightarrow \exists y\,(y^2=x))$

   注意：量词顺序很重要，$\exists y$ 必须在 $\forall x$ 之后。

4. 假设素数有限，设为 $p_1, p_2, \ldots, p_n$。构造 $N = p_1 p_2 \cdots p_n + 1$。$N$ 不能被任何 p_i 整除（余数都是 1），所以 N 要么是新的素数，要么有新的素因子——矛盾。

5 补充阅读

推荐材料

- **Background Reading** —Keith Devlin 撰写的课程背景介绍，涵盖数学的历史发展、数学符号的必要性、以及数学思维的价值。
- **Set Theory Supplement** —集合论基础速查表，包括集合表示法、空集、子集、并集、交集等核心概念。后续课程会频繁使用这些记号。